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Alternative explanations to the differences of femoral and brachial saline contrast injections for echocardiographic detection of patent foramen ovale

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Summary Patent foramen ovale (PFO) of the interatrial septum is a cardiac foetal remnant, which frequent persistence in adulthood has important implications in a variety of clinical conditions. Echographic diagnosis of PFO is based on detection of interatrial shunt by means of contrast microbubbles identification after venous injection of a first-generation echographic contrast agent. Current recommendations propose venous femoral injection of contrast for enhanced echographic detection of PFO instead of venous brachial administration, as femoral injection has been shown to have higher sensitivity for PFO detection. Inferior vena cava inflow directed toward interatrial septum has been considered the explanation for increased sensitivity of femoral delivery of contrast. In the present paper, it is hypothesised that the main determinants of these differences between injection sites are technical factors related to right atrial contrast opacification and proper transient right atrial pressure rise, rather than intraatrial flow streaming. Effects of inferior vena cava inflow stream, although significant during foetal life, would be negligible after birth. Rationale and evidence, basis for further research, and practical implications leading to a simpler and safer routine technique for echographic detection of PFO are presented and discussed.

Introduction

In foetal circulation the foramen ovale (FO) is necessary for oxygenated blood from inferior vena cava (IVC) to cross the interatrial septum (IAS) and bypass pulmonary circulation. IVC flow streams preferentially toward IAS and FO, while superior vena cava (SVC) and coronary sinus flows are directed away from IAS and FO [1]. Circulatory changes at birth rise left atrial pressure, making septum primum flap to close against septum secundum, abolishing right to

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left shunt through FO [1,2]. True fibrous anatomical fusion of primum and secundum septa usually occurs early in life, but in 25—30% of cases fusion is incomplete, and patent foramen ovale (PFO) persists as development remnant [3].

PFO is clinically relevant since it has been related to stroke [4], migraine [5], decompression illness [6] and platypnoea-orthodeoxya syndrome [7]. Its identification is also valuable for some non-cardiac surgical procedures [8]. Diagnosis relies upon echographic detection of microbubbles in left atrium or systemic arterial circulation shortly after venous administration of a first-generation contrast agent [9] that is cleared from circulation after passage through a pulmonary or systemic capillary bed. PFO screening accounts for a substantial portion of contrast echocardiography studies [10] that is presumed to increase if causative relation with the above mentioned clinical conditions is confirmed and therapeutic interventions are spread [11].

Transoesophageal echocardiography (TOE) has been considered for some time the method of choice in the diagnosis owing to its superiority over fundamental imaging transthoracic echocardiography (TTE) [12], but at the present time, harmonic imaging TTE has been shown to be more sensible than TOE [13]. Transcranial Doppler sonography (TDS) [14] is used clinically as well, and transmitral Doppler transthoracic echocardiography [15] has also been studied in this setting. Besides imaging modalities, other issues are relevant concerning PFO detection. Several solutions have been used as contrast agents, although agitated saline contrast (ASC) is the most spread [16]. Right atrial pressure need to be greater than left atrial pressure for right-to-left shunt to occur. Valsalva or cough manoeuvre should be performed systematically to reach higher sensitivity [17]. For PFO exclusion, transient leftward shifting of the interatrial septum should be demonstrated to ensure that right atrial pressure has raised above left atrial pressure after right atrial contrast opacification [9].

Venous femoral injection of saline contrast has been demonstrated to be superior to more usual and easy brachial injection for PFO detection [18,19]. Current state-of-the-art reviews recommend femoral or pedal vein saline injections for optimal detection of PFO, all invoking the effect of streaming of inferior vena cava inflow, Eustachian valve contribution, and FO oblique slit-like anatomical conformation [9,16,20].

Hypothesis

Enhanced echographic detection of PFO after femoral intravenous injection of agitated saline may be mainly determined by denser right atrial opacification after central venous infusion due to shorter injection time and faster arrival to heart, allowing a lesser dilution of contrast. Inferior vena cava inflow streaming and foetal remnant anatomic structures would play a minor role (if any) in adulthood. This hypothesis does not refuse the importance of streaming in foetal circulation, and does not deal with the clinical implications of PFO persistence after birth. The streaming effect accepted in the adult has been an attractive hypothesis, but optimisation of PFO detection should rather address to the accomplishment of complete right atrial opacification, and to the performance of manoeuvres to achieve proper transient rise of right atrial pressure.

Evidence and discussion

Current recommendations are based mainly in two studies [18,19] that compared femoral and brachial contrast injection for PFO detection. The first one [18] compared brachial and femoral administration of 10 ml dextrose contrast for shunt detection by means of transthoracic echocardiography. Femoral injection was shown to be more sensible than brachial injection for PFO detection. Positive studies were semiquantitatively graded for left atrial opacification, which was found to be significantly greater during femoral contrast delivery than during upper extremity contrast injection. Satisfactory right heart opacification was reported for both injection modalities, but quantitation was not undertaken and its potential contribution to the global results was not evaluated. The second study [19] evaluated patients undergoing TEE and TDS simultaneously. Femoral and brachial injections of a first-generation echocontrast were sequentially compared for PFO detection with both imaging modalities. Femoral injection was found superior for PFO detection. TDS showed higher amount of contrast in intracranial arteries after femoral injection. This fact was attributed to increased shunt volume after femoral injection, but atrial contrast was not quantified in the TEE study. Of note, the brain transit time was shorter for femoral injection than for antecubital injection. In their early work about contrast echocardiography, Seward et al. [21] found that most superficial veins were suitable for contrast injection and produced studies ''similar'' in quality compared to caval injections for a variety of clinical purposes, but the grade of right atrial opacification was not specifically assessed.

Recent recommendations [9] credit the advantage of venous femoral injection for PFO detection, but state that essential requirements for PFO detection and exclusion are dense right atrial opacification and interatrial septal shift to the left, as evidence of right atrial pressure exceeding that of left atrium. That seems inconsonant with available evidence, first because streaming of contrast in right atrium has not been studied by direct visualization, and second because the slightest effect of intraatrial streaming would become negligible if high atrial pressure were achieved after homogeneous dense opacification of right chambers. Of interest, these recommendations themselves acknowledge that intravenous lines in the feet have low sensitivity for PFO detection owing to long venous transit times and contrast bubbles destruction.

Paradoxical embolism from superior vena cava though a PFO has been clearly documented [22], showing how flow across PFO is mainly dependent of pressure gradient in adulthood, and educing a corrected appraisal of caval inflow role. Flow from superior vena cava shunting though a PFO causing platypnoea-orthodeoxya syndrome has also been demonstrated [23].

The results of the studies [18,19] above mentioned could have been determined by enhanced right atrial opacification after contrast injection in a central vein with fast and direct flow to the right heart, rather than by the angulation of flow of inferior vena cava. Future studies addressing the meaning of right intraatrial flow angulation and streaming should carefully control intensity and circulation rate of contrast injected in several central and peripheral veins when arriving to right atrium. Contrast technique [16] and Valsalva manoeuvre [17] should be controlled. Such investigation may need several venous punctures, and test different sized needles or venous lines. Such kind of protocol poses ethical concerns as venous canalization is not free of complications [24,25]. Development of new echocardiographic tools [26] may offer non-invasive estimation of intraatrial pressure gradients in the future to clarify the actual weight of flow streaming in interatrial PFO-related shunt in the adult.

Practical implications

Echographic studies committed to PFO detection seem to increase in number and proportion. Current recommendations are founded on attractive assumptions based on foetal physiology, but lacking proper demonstration in adults. Proper right atrial opacification with optimal contrast infusion through a peripheral antecubital vein and effective transient rise in right atrial pressure may suffice for excellent sensitivity of PFO detection, thus avoiding the significant increase in procedure technical complexity and patient risks [24,25].

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